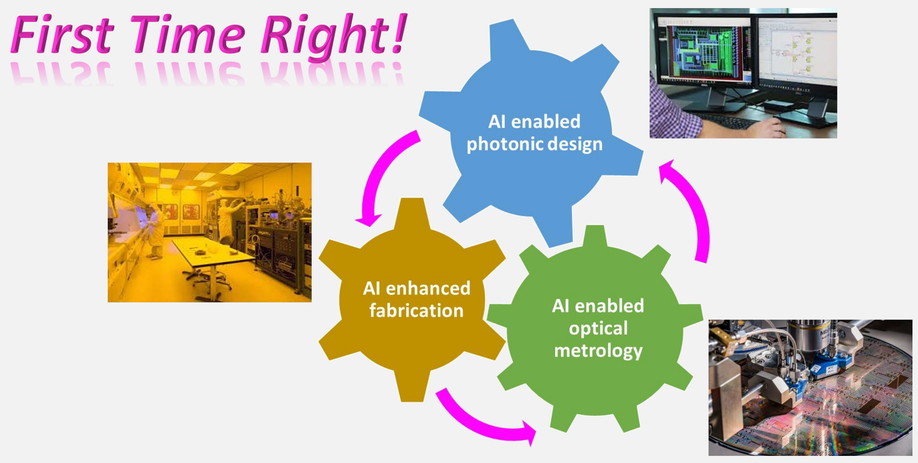

AI-Driven Integrated Nanophotonics: Inverse Design ↔ Fabrication ↔ Metrology

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Contributors: Martin Vachon[1], Shurui Wang[1], Muhammad Al-Digeil[2]

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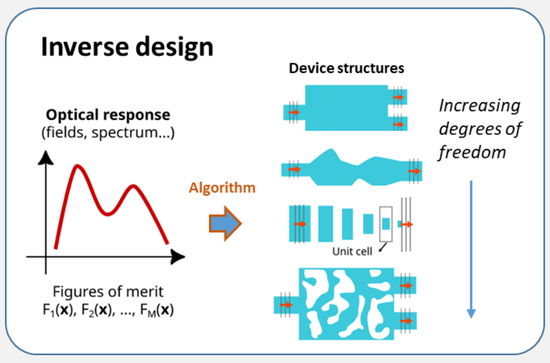
AI assisted inverse design

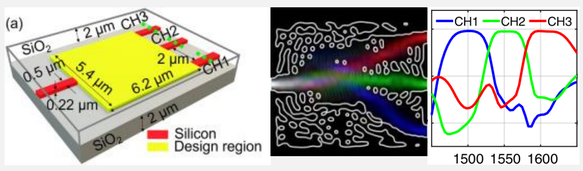
McGill
Prof. O. Liboiron-Ladouceur
G. Zhang, M. M. Masnad, A. Li, R. Rogers, L. Pascar

High-throughput and Secure Networks Challenge program
Artificial Intelligence for Design Challenge program

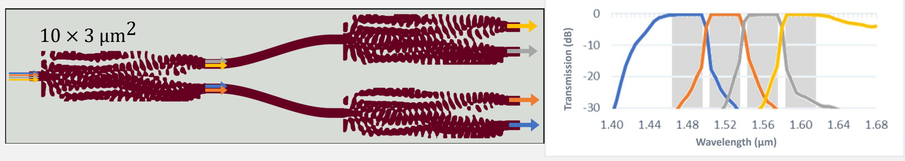
- Ultra-compact devices
- Advanced functionality
- Intricate, non-intuitive geometries
- Robustness via energy constraints
- Integrates thermo-optic effects



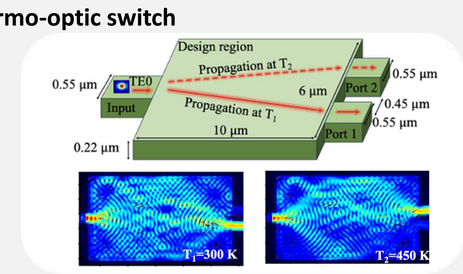
WDM: optimization with energy constraint



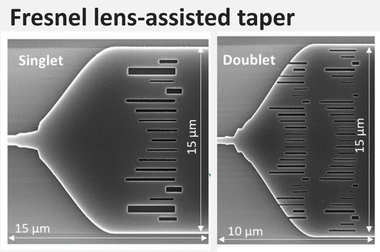
4-channel WDM cascaded filters



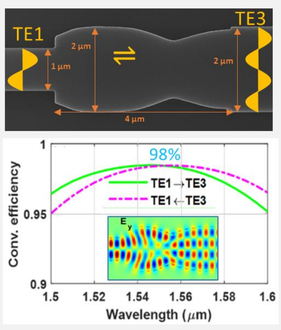
Thermo-optic switch



Fresnel lens-assisted taper



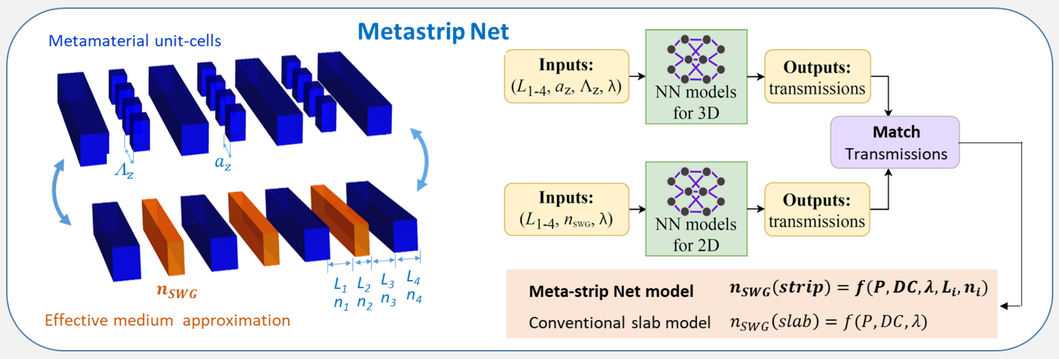
Mode converters



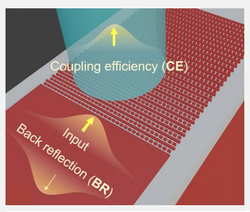
uOttawa
Prof. L. Ramunno
Q. Wang, N. S. Israel, S. Patel, D. Blankespoor

Artificial Intelligence for Design Challenge program

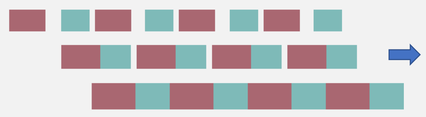
- **Meta-strip Net: enabling pseudo-3D simulation**
- **Extends effective-medium approximation to complex unit cells using an *environment-aware* surrogate model**
- **Accuracy of a 3D simulation at the compute cost of 2D**



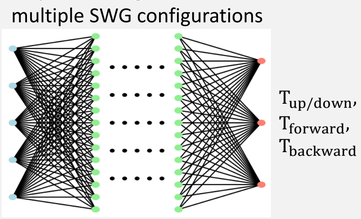
Applied to the inverse design of ***perfectly vertical*** **grating couplers**:
2D FDTD + Meta-strip Net instead of 3D FDTD ⇨ x 400 speedup

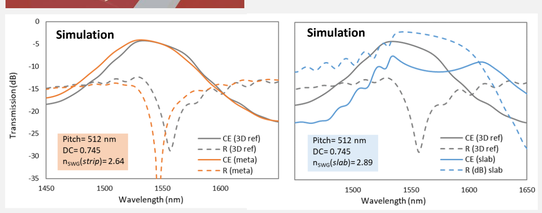


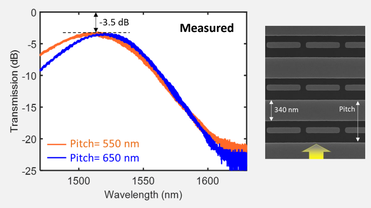
Other SWG configurations



Deep NN surrogate models for multiple SWG configurations







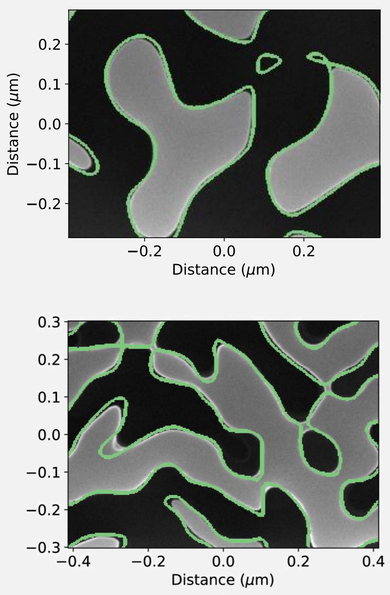
Digital fabrication twin

McGill
Prof. O. Liboiron-Ladouceur
D. Gostimirovic, A. Li, R. Azimi

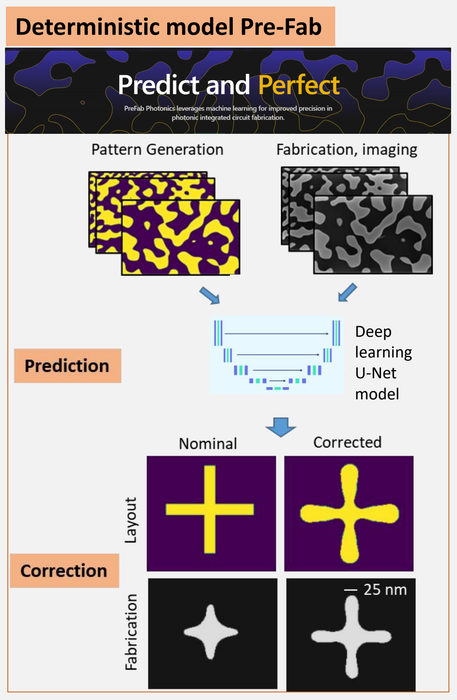
Artificial Intelligence for Design Challenge program

- **Reliable fabrication of small features is critical for miniaturization**
- **AI-based digital twin predicts spatially varying fabrication deviations**
- **Enables virtual design validation, adaptive layout correction, fabrication-aware design**

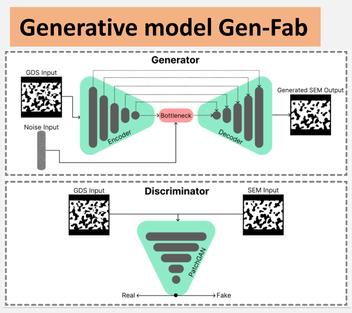
Design vs Reality (SEM)



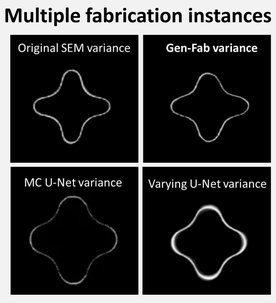
Deterministic model Pre-Fab



Generative model Gen-Fab



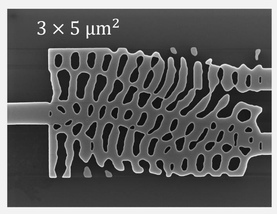
Multiple fabrication instances



- Gen-Fab provides realistic predictions of fabrication variations
- Enables robustness optimization

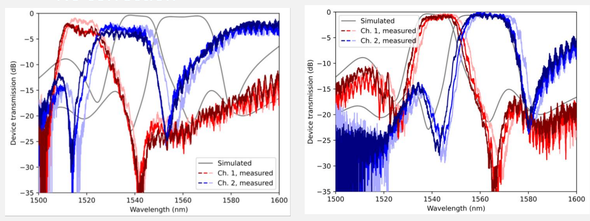
Compact CWDM

- $3 \times 5\ \mu m^2$ footprint
- 20 nm channel spacing with 10 nm flat passband
- Among the *narrowest* channel spacings reported for inverse-designed WDM devices
- Performance approaches simulation and is on par with compact classical designs (Echelle, Contra-DC, MZI) at much smaller footprint

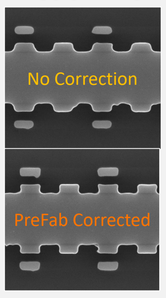


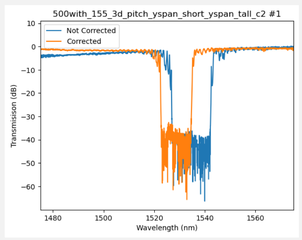
- Performance without and with Prefab correction
  - Loss 4 dB → < 1 dB
  - Channel shift 23 nm → 3.8 nm
  - Crosstalk -14 dB → -22 dB

No correction — Prefab corrected



Bragg grating





- Trained on irregular geometries emulating topology-optimized devices
- Generalizes directly beyond the training regime to:
  - Manhattan-like layouts
  - Classical photonic devices

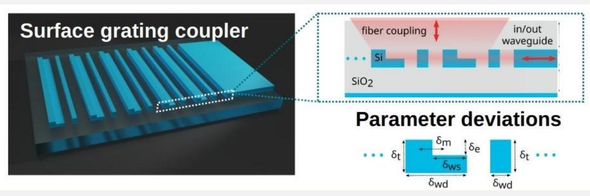
- Corner sharpening
- Bandwidth narrowing : 18 → 13 nm
- Repositioning of λ_0 : 1534 → 1529 nm

AI-enabled optical metrology

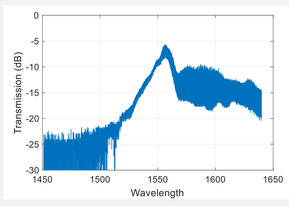
Prof. D. Melati
A. Masominia
M. Gupta (NRC co-op)

Artificial Intelligence for Design Challenge program

- **Retrieve structural parameters without needing imaging**

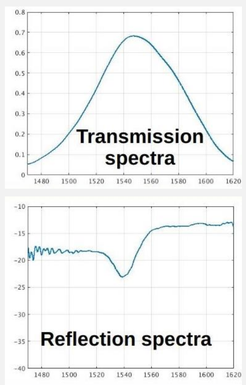


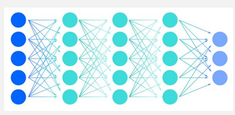
- Nanophotonic fabrication introduces small dimensional variations that significantly affect device performance
- Efficient, accurate metrology is highly desirable

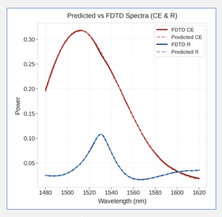


- Optical response intrinsically encodes structural information

Fourier transform







- Using grating architectures, ML-based inversion enables high-precision retrieval of complex unit-cell geometry, including deeply subwavelength features
- Beyond conventional ring/MZI metrology limited to ~450–500 nm waveguide widths
- Extendable to material refractive index characterization
- Non-destructive, wafer-scale process monitoring

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for further information

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